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Model-based deep learning enables time-resolved computational microscopy

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Abstract

Computational microscopy combines advances in optical hardware and signal processing to push the boundaries of imaging resolution and functionality. However, acquiring extended information often comes at the expense of temporal resolution. Here, we present a model-based deep learning framework for time-resolved imaging in multi-shot computational microscopy. Building upon the plug-and-play (PnP) optimization theory, our approach integrates the low-level spatiotemporal priors learned from large-scale video datasets with the physical model of an optimized measurement scheme, enabling accurate, time-resolved reconstruction of dynamic scenes. Using lensless coded ptychographic microscopy as an example, we experimentally demonstrate high-speed holographic imaging of an order of magnitude faster sample dynamics without compromising quality. Additionally, we show that the proposed framework enables high-throughput, label-free imaging of various biological activities of freely moving organisms, such as paramecia and rotifers, with a sensor-limited space-bandwidth-time product of 227 megapixels per second. The presented approach provides a promising solution to time-resolved computational microscopy across a broad range of imaging modalities.

Introduction

Computational microscopy leverages advancements in both optical hardware and signal processing techniques to achieve enhanced resolution [1–4], extended modalities [5–8], improved depth penetration [9–11], and volumetric imaging [12–14], etc. Due to the information throughput limit of optical systems or image sensors, computational microscopy often relies on sequential measurements, each of which encodes partial information of the higher dimensional data to be measured. The requirement of multi-shot acquisition inevitably compromises temporal resolution, posing practical challenges for capturing fast dynamics.

To address this limitation, parallel acquisition schemes have been proposed [15–21], where multiple encoded images are obtained simultaneously using an array of imaging modules. However, the increased system complexity and substantial cost render this approach impractical for certain applications. Alternatively, multi-shot measurements can also be compressed into a single capture by means of multiplexing [22–32], albeit at the cost of imaging quality or imposing additional constraints on the samples.

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The recent advancements in deep learning have revolutionized computational microscopy across a wide range of modalities [33–38]. By incorporating diverse forms of data priors, these methods help mitigate the ill-posed nature of the inverse problem, enabling competitive image reconstruction from fewer measurements or even a single shot [39–50]. Despite their remarkable success, deep learning-based image reconstruction methods are now approaching the information-theoretic limits of imaging [51]. The finite amount of information contained in a single measurement inevitably leads to a trade-off between reconstruction accuracy and generalizability across diverse sample types.

More recently, growing attention has been devoted to exploiting the spatiotemporal correlations inherent in dynamic scenes to further enhance imaging performance [52–63]. Inspired by these advances, in this work, we present a model-based deep learning framework that leverages spatiotemporal data priors to overcome this challenge and achieve time-resolved imaging in multi-shot computational microscopy. Building upon the plug-and-play (PnP) optimization theory, our approach integrates the physical forward model of the imaging system with a state-of-the-art deep video denoiser serving as an implicit regularizer. Trained on large-scale natural-scene video datasets, the denoising network captures the intrinsic low-level spatiotemporal features shared across diverse dynamic processes, thereby enabling cross-domain knowledge transfer without requiring domain-specific data priors. Combined with an optimally designed multi-shot acquisition scheme based on coded ptychography, our framework experimentally achieves lensless holographic microscopy with an order-of-magnitude improvement in imaging speed, reaching a temporal resolution of 112 fps and an information throughput of 227 megapixels per second. We further validate the fidelity and generalizability of the method by imaging various biological activities of freely moving organisms, including paramecia and rotifers, demonstrating physically reliable reconstructions at single-shot temporal resolution.

Results

High-speed lensless microscopy based on coded ptychography

Lensless microscopy is a holographic imaging modality that features unique advantages in terms of high throughput, compact design, and cost-effectiveness [64–68]. It has found diverse applications including digital pathology [69, 70], endoscopy [71–73], live cell imaging [74–76], polarization microscopy [77, 78], environmental monitoring [79], and point-of-care testing [80–82]. In a typical lensless microscope, optical phase information of the sample is obtained through diversity measurement, which is commonly implemented by varying the illumination wavelength [83, 84], imaging distance [85, 86], or wavefront modulation [87]. In this work, modulation diversity is adopted due to its superior information encoding efficiency (Supplementary Note 1), and a proof-of-concept lensless microscope was developed based on the principle of coded ptychography [88].

The experimental setup is conceptually illustrated in Fig. 1a and detailed in Supplementary Fig. 1. The sample is illuminated by a collimated coherent beam from a fiber-coupled laser. The exit wave is then modulated by a thin diffuser placed a few millimeters downstream from the sample, and the intensity of the resulting speckle field is captured by a board-level image sensor. To realize diversity measurement, the sample is laterally translated to different scanning positions and the corresponding modulated intensity

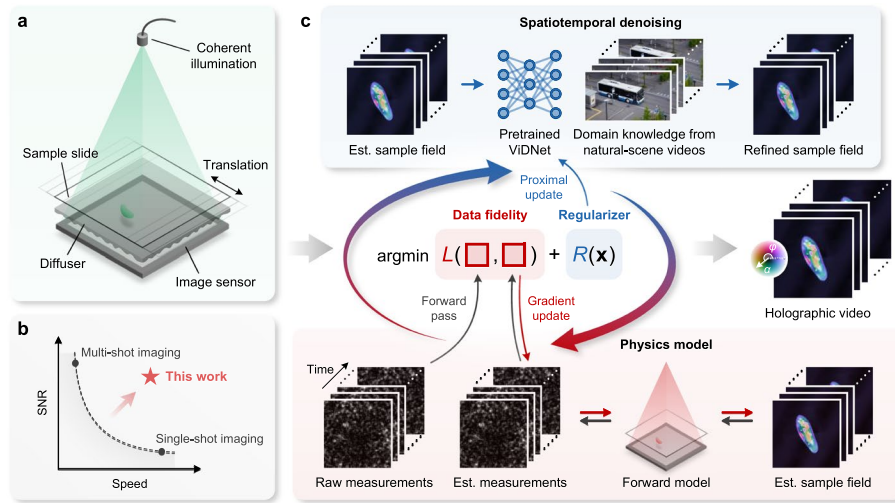


Fig. 1 Overview of the model-based deep learning framework. **a** Schematic of the lensless coded ptychographic microscope. During measurement, the sample is laterally translated and intensity images are captured sequentially for holographic reconstruction. **b** The inherent tradeoff between imaging speed and signal-to-noise ratio (SNR) faced by conventional multi-shot imaging systems. This work aims to improve the imaging quality without compromising temporal resolution by exploiting deep spatiotemporal priors. **c** Illustration of the model-based deep learning reconstruction framework. The algorithm proceeds by iteratively updating with respect to the physics model and the spatiotemporal prior and converges to a time-resolved holographic video. The denoising network is trained on a natural-scene video dataset, transferring low-level spatiotemporal features to holographic video enhancement

images are acquired. Due to the lensless configuration, the spatial resolution is primarily limited by the sensor pixel size, which is $2.74 \mu\text{m}$ in our case, although it can be potentially improved toward the diffraction limit with proper modeling of the pixel's intensity response [89]. The image sensor has a resolution of $2,848 \times 2,848$, corresponding to a field of view (FOV) of $7.8 \times 7.8 \text{ mm}^2$. The exposure time was approximately 0.1 ms so that the measurement can be performed in fly-scan mode, with the sensor operating at its maximum framerate.

Based on the lensless imaging platform, the forward model can be expressed as

$$\mathbf{y}_k = |\mathbf{Q}(d_2)\text{diag}(\mathbf{m})\mathbf{Q}(d_1)\text{diag}(\mathbf{p})\mathbf{T}(\mathbf{r}_k)\mathbf{x}_k| = |\mathbf{A}_k\mathbf{x}_k|, \quad (1)$$

where $\mathbf{y}_k$ and $\mathbf{x}_k$ denote the captured field amplitude and the underlying sample field with respect to the k th time point, respectively. $\mathbf{T}(\mathbf{r})$ and $\mathbf{Q}(d)$ denote the lateral translation and free-space propagation operators parameterized by a shift vector $\mathbf{r}$ and a propagation distance d , respectively. $\mathbf{m}$ and $\mathbf{p}$ denote the diffuser and illumination probe profiles, respectively. The linear measurement operators can also be compactly represented as an abstract measurement matrix $\mathbf{A}_k$. Accurate modeling of the imaging process requires precise knowledge on multiple system parameters, including the translation position at each time point $\mathbf{r}_k$, the sample-to-diffuser distance d_1 , the diffuser-to-sensor distance d_2 , the diffuser modulation $\mathbf{m}$ and the probe field $\mathbf{p}$, all of which are obtained through calibration experiments or data preprocessing. To further compensate for experimental imperfections, an additional algorithmic self-calibration scheme is introduced, which jointly refines the sample function and system parameters by exploiting the information redundancy of the measurement data (Methods).

Deep plug-and-play optimization for time-resolved holographic reconstruction

For a multi-shot imaging system, given the sequential measurements $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K$, recovery of the time-dependent sample field $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K)$ can be realized by solving the following inverse problem:

$$\hat{\mathbf{x}} = \underset{\mathbf{x}}{\operatorname{argmin}} \{F(\mathbf{x}) + R(\mathbf{x})\} \quad \text{with} \quad F(\mathbf{x}) = \sum_{k=1}^K L(|\mathbf{A}_k \mathbf{x}_k|, \mathbf{y}_k), \quad (2)$$

where F and R denote the data-fidelity function and the regularization function, respectively. The data-fidelity function ensures that the estimated sample field strictly adheres to the forward model given by Eq. (1). The ℓ_2 loss with respect to the field amplitude is chosen as the error metric L in this work, although it can take other forms depending on the noise statistics. Given the high ill-posedness of the inversion task, spatiotemporal priors of the sample field should be incorporated as a means of regularization. The regularizer R can appear in explicit forms that enforce hard constraints [90–92] or promote low-rank [93] or sparse [57, 94, 95] solutions. However, these hand-crafted priors are sometimes too simple and heuristic to fully capture the complex spatiotemporal features, resulting in limited reconstruction capability of dense and fast-moving samples. Previous studies have also proposed warm-start strategies that use reconstructions at previous time points as initial estimates, reducing the number of required measurements and thereby improving temporal resolution [96]. Nevertheless, achieving single-shot temporal resolution remains challenging.

Plug-and-play methods provide a promising solution that elegantly integrates physical models and data priors learned by deep neural networks [97, 98]. To elucidate the concept of PnP, we first revisit the classical proximal gradient method for solving Eq. (2):

$$\mathbf{u}^{(i)} = \mathbf{x}^{(i-1)} - \gamma \nabla_{\mathbf{x}} F(\mathbf{x}^{(i-1)}) \quad (3)$$

$$\mathbf{v}^{(i)} = \operatorname{prox}_{\gamma R}(\mathbf{u}^{(i)}) \quad (4)$$

$$\mathbf{x}^{(i)} = \mathbf{v}^{(i)} + \beta^{(i)}(\mathbf{v}^{(i)} - \mathbf{v}^{(i-1)}), \quad (5)$$

where $\mathbf{x}^{(0)} = \mathbf{v}^{(0)}$ is the initial estimate, i denotes the iteration number, and γ denotes the step size. $\beta^{(i)}$ is a parameter controlling the extrapolation step, and when $\beta^{(i)} = i/(i+3)$, Eqs. (3)–(5) are also known as the fast iterative shrinkage thresholding algorithm (FISTA) for linear inverse problems [99]. The update step with respect to the regularizer R relies on its proximal operator [100]:

$$\operatorname{prox}_{\gamma R}(\mathbf{u}) = \underset{\mathbf{x}}{\operatorname{argmin}} \left\{ \frac{1}{2} \|\mathbf{x} - \mathbf{u}\|_2^2 + \gamma R(\mathbf{x}) \right\}. \quad (6)$$

The above formulation can be interpreted as a maximum a posteriori (MAP) estimation of a video denoising problem under additive white Gaussian noise (AWGN), where $\mathbf{u}$ denotes the noisy observation, γ denotes the noise variance, and $R(\mathbf{x}) = -\ln p(\mathbf{x})$ represents the data prior (Supplementary Note 2). This perspective indicates the

possibility of replacing the analytical proximal operator with a black-box AWGN denoiser: $\mathbf{v}^{(i)} = \mathcal{D}_\sigma(\mathbf{u}^{(i)})$, where $\sigma > 0$ controls the relative strength of the denoising operator $\mathcal{D}$. As a result, advanced spatiotemporal data priors can be readily incorporated into the iterative process to enhance the reconstruction quality without compromising physical reliability, thereby breaking the trade-off between imaging quality and speed for multi-shot computational microscopy, as shown in Fig. 1b.

To learn the complex spatiotemporal features of dynamic scenes, we developed and trained a non-blind AWGN video denoising network (ViDNet). Compared with off-the-shelf video denoisers such as FastDVDnet [101], ViDNet has been specifically tailored for PnP-based video restoration tasks [102–105]. First, a deeper network architecture is adopted to capture long-range spatiotemporal features. This is proven to be beneficial for maintaining finer spatiotemporal textures, enabling reconstruction of approximately an order of magnitude faster dynamics while maintaining competitive quality (Supplementary Notes 3 and 4). Second, the entire network is bias-free, which enables better scalability to unseen noise levels and ensures stable convergence during PnP iterations, even under challenging conditions involving nonlinear forward models, accelerated algorithms, or poor initialization (Supplementary Note 4). Upon finishing training, ViDNet, together with a three-dimensional total variation (3DTV) regularized video denoising step, is invoked iteratively to enforce prior knowledge in the PnP optimization process (Methods), yielding significantly improved performance over state-of-the-art spatiotemporal PnP priors on dynamic scene recovery (Supplementary Note 4). The entire reconstruction pipeline is conceptually illustrated in Fig. 1c.

It is worth noticing that neural network-based algorithms for multi-shot dynamic scene recovery have made remarkable advancements in recent years. Compared with algorithms based on untrained decoder networks [53, 106, 107] or implicit neural representation [56, 59], our approach features two distinct advantages. First, the pretrained ViDNet can implicitly handle different types and degrees of motions by learning from large-scale video datasets, whereas previous approaches rely on no data priors but specific network architectures to model sample dynamics, leading to scene-dependent imaging performance. Second, the PnP algorithms are more computationally efficient because only a few times of network inference is required during the reconstruction, obviating the need of test-time training from scratch.

One major practical challenge in integrating deep learning into computational imaging modalities is the experimental difficulty, or even infeasibility, of collecting clean and labeled training datasets, particularly for dynamic video sequences. To address this, we leverage widely available video datasets of natural dynamic scenes from the computer vision community for network training. We emphasize that this cross-modality knowledge transfer is made possible owing to two reasons. First, although the training data and the imaging samples differ significantly in high-level semantic contents, they share strong similarities in low-level spatiotemporal features, such as edges, textures and patterns, which are inherent to natural dynamic processes. Second, the judiciously designed measurement scheme achieves relatively high information encoding efficiency, as quantified by Fisher information analysis. This significantly reduces measurement uncertainty at the hardware level, thereby suppressing the hallucination artifacts from neural networks (Supplementary Note 1 and Supplementary Fig. 2).

Experimental validation and characterization

We first collected the coded ptychographic dataset of a static xylophyta dicotyledonous stem sample in transversal section, as shown in Fig. 2a. To investigate the reconstruction quality under varying degrees of motion, an offset value is intentionally added to the calibrated sample position at each time point during the reconstruction to simulate a global translation motion. For comparison, we also implemented PnP-FISTA with TV regularization as well as DRUNet and FastDVDnet, one of the state-of-the-art PnP priors for image and video restoration, respectively [108]. Different sample translation speeds ranging from 0 to 9 pixels/frame were evaluated, with $K = 10$ images used for reconstruction. The amplitude peak signal-to-noise ratio (PSNR) is introduced as a quantitative metric, which is calculated using a reference image reconstructed from 100 ptychographic measurements. We also registered the reconstructed frames and calculated the inter-frame correlation coefficients to quantify the temporal consistency. The results are presented in Fig. 2b, c and Supplementary Video 1. As the sample speed increases, the quality of the 3DTV-based reconstruction drops drastically toward that of the frame-by-frame baseline method where no temporal priors are used. DRUNet and FastDVDnet can generate more natural-looking textures but the results tend to overfit to the data priors and deviate from the ground truth due to a lack of long-range spatiotemporal correlations. In contrast, the proposed ViDNet-based algorithm maintains fine spatial details and a high temporal consistency even under large sample motions, with a PSNR improvement by over 2 dB. It is also noteworthy that for the baseline PnP-3DTV and PnP-DRUNet+3DTV algorithms, the parameters of the spatiotemporal TV regularizer has been optimized for each sample speed, whereas PnP-ViDNet+3DTV adopts the same algorithm parameters in all test cases, demonstrating its superior generalizability to various degrees of motion. Additional experimental results on different samples, including a section of the *Parascaris equorum* uterus and tissue slices, are provided in Supplementary Note 4.

The imaging resolution and phase reconstruction accuracy were also evaluated using a quantitative phase target (QPT, Benchmark Technologies), as shown in Fig. 2d-f. In all three test cases with varying translation speeds, Group 7, Element 4 in the 1951 USAF chart is resolved, indicating that the spatial resolution maintains at the sensor pixel's Nyquist sampling limit. The reconstructed phase profiles also remain in good agreement with the nominal values.

Dynamic holographic microscopy of freely moving organisms

To investigate the applicability of PnP-ViDNet on real-world dynamic scene recovery, we prepared and conducted imaging of various living biological samples. A smaller region of interest of $1,424 \times 1,424$ pixels was selected from the sensor FOV, allowing for a higher acquisition framerate of 112 fps. Such configuration corresponds to a space-bandwidth-time product of ~ 227 megapixels per second, which is primarily limited by the sensor's data transmission bandwidth.

We first validate the effectiveness of algorithmic calibration (Fig. 3a and Supplementary Video 2) and compare different spatiotemporal priors by imaging a fast-moving paramecium (Fig. 3b and Supplementary Video 3). The direct reconstruction exhibits background artifacts, which are effectively suppressed through algorithmic

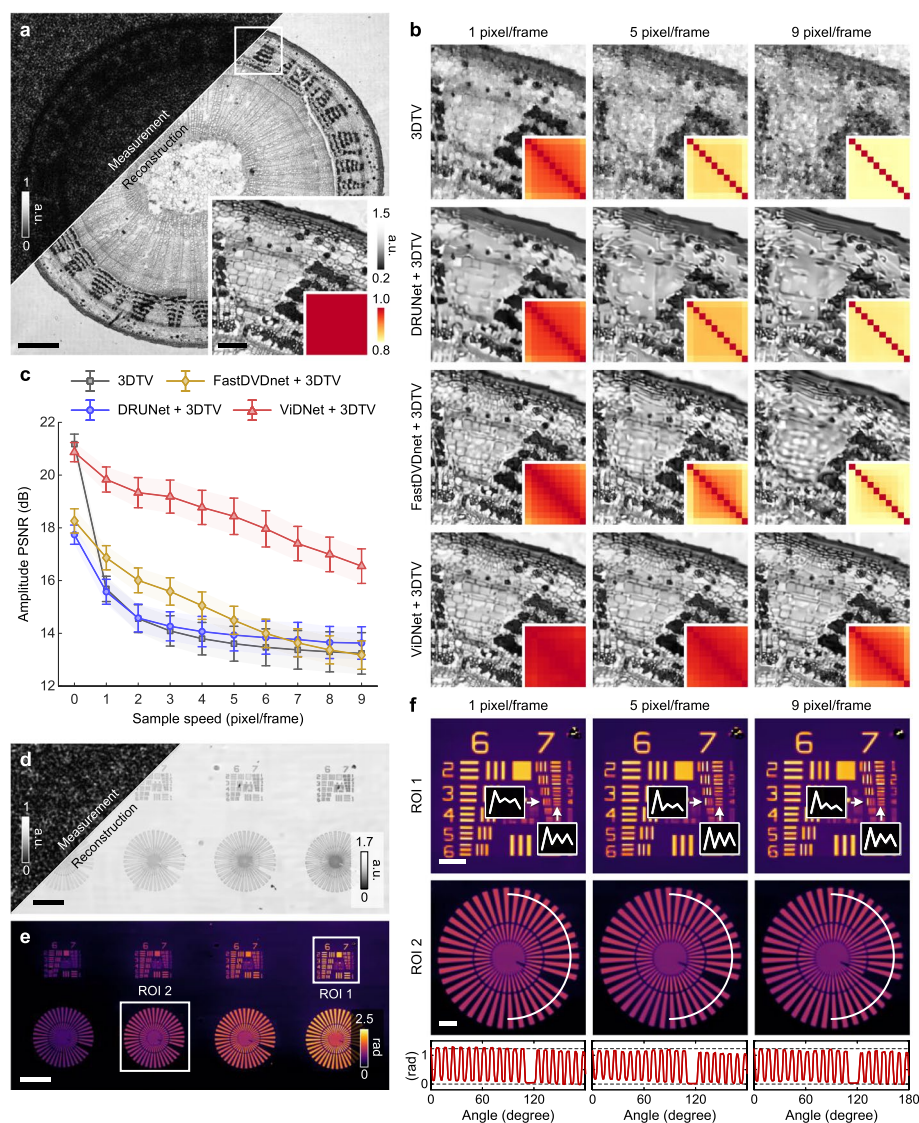


Fig. 2 Quantitative evaluation based on experimental data. **a** Representative raw intensity image and the reconstructed amplitude of a static xylophyta dicotyledonous stem sample using 100 diversity measurements. The inset shows the enlarged region of interest (ROI) and the inter-frame correlation matrix. **b** Reconstructed amplitude and the inter-frame correlation matrix from $K = 10$ diversity measurements under different translation speeds (Supplementary Video 1). **c** Amplitude PSNR values as a function of the sample speed, calculated over 100 ROIs. Error bars represent the standard deviation. **d** Representative raw intensity image and the reconstructed amplitude of a quantitative phase target by PnP-ViDNet+3DTV. **e** Reconstructed phase of the target by PnP-ViDNet+3DTV. **f** Enlarged views of two ROIs under different sample speeds. The cross-sectional profiles of the smallest resolvable line group and the focus star pattern are shown, where the dashed lines indicate the ground-truth values. Scale bars, 500 μm (a), 100 μm (inset in (a)), 200 μm (d, e), 50 μm (f)

self-calibration. It can be deduced that these artifacts mainly arise from the inaccuracies of the diffuser and probe modeling, as they appear as a fixed pattern with a time-varying displacement along the sample translation direction. Improved visual quality is also observed from the ViDNet+3DTV reconstruction over the one based purely on 3DTV. Samples at a high speed often violate the temporal gradient-domain sparsity assumption,

and thus 3DTV-based reconstruction tends to result in blurring effects along the moving direction. In contrast, ViDNet is able to extract long-term spatiotemporal correlations and preserve finer subcellular details, such as the cell membrane around the vacuolar structures as highlighted by the cross-sectional profiles P1 and P2.

Compared with brightfield microscopy, lensless microscopy offers an additional phase channel to enhance the contrast of various biological components. Essentially, it quantifies their morphological properties and refractive indices, both of which contribute to the optical path length variations that determine the phase retardance. For example, while food vacuoles are optically transparent structures, they exhibit distinct phase values due to their refractive index contrast with the surrounding cytosol. Resolving and identifying these subcellular components enables easier tracking of cell behaviors, and its rotation movements can be clearly visualized (Fig. 3c-h and Supplementary Video 4). More interestingly, benefiting from the time-resolved imaging capability, we experimentally observed fast biological activities such as the formation and disappearance of food vacuoles (Fig. 3i-m and Supplementary Video 5). During formation, the vacuolar membrane encloses and separates from the surrounding cytoplasm. Upon complete digestion, the vacuole gradually decreases in size before finally disappearing. Both processes are recorded at a high temporal resolution and with clear visual contrast.

Figure 4 presents a qualitative comparison of different spatiotemporal PnP priors on two fast-moving paramecium samples. In the first case (Fig. 4a, b, and Supplementary Video 6), the sample continuously moves in the same direction at a high speed of approximately 1 mm/s. Reconstructions based on 3DTV, DRUNet, and FastDVDnet exhibit varying degrees of temporal fluctuation due to their limited ability to capture long-range spatiotemporal dependencies, whereas the ViDNet-based prior achieves stable video recovery. In the second case (Fig. 4c, d, and Supplementary Video 7), the sample exhibits abrupt stops and reversals in direction. Such sudden motion changes lead to noticeable striped artifacts in the spatiotemporal slices reconstructed by 3DTV, DRUNet, and FastDVDnet. In contrast, ViDNet maintains much higher temporal consistency. These results demonstrate that ViDNet significantly outperforms existing PnP priors in reconstructing fast and complex motions.

To further demonstrate the dynamic imaging capability, samples containing live rotifers and paramecia were imaged, as shown in Fig. 5 and Supplementary Video 8. Tens of freely moving organisms within a relatively large FOV of $3.9 \times 3.9 \text{ mm}^2$ are monitored simultaneously. In contrast to paramecia, rotifers exhibit slower locomotion but demonstrate more complex environmental interactions, including body extension and contraction. They may also appear in dense colonies, and the time-resolved reconstruction is shown in Fig. 5i and Supplementary Video 9. The retrieved sample field contains complicated spatial textures, which are typically considered challenging for conventional reconstruction algorithms. As an additional benefit of holographic imaging, the sample-to-sensor distance can be adjusted flexibly during post processing, enabling the reconstruction of in-focus sample images at different axial locations, as highlighted in Fig. 5l, m.

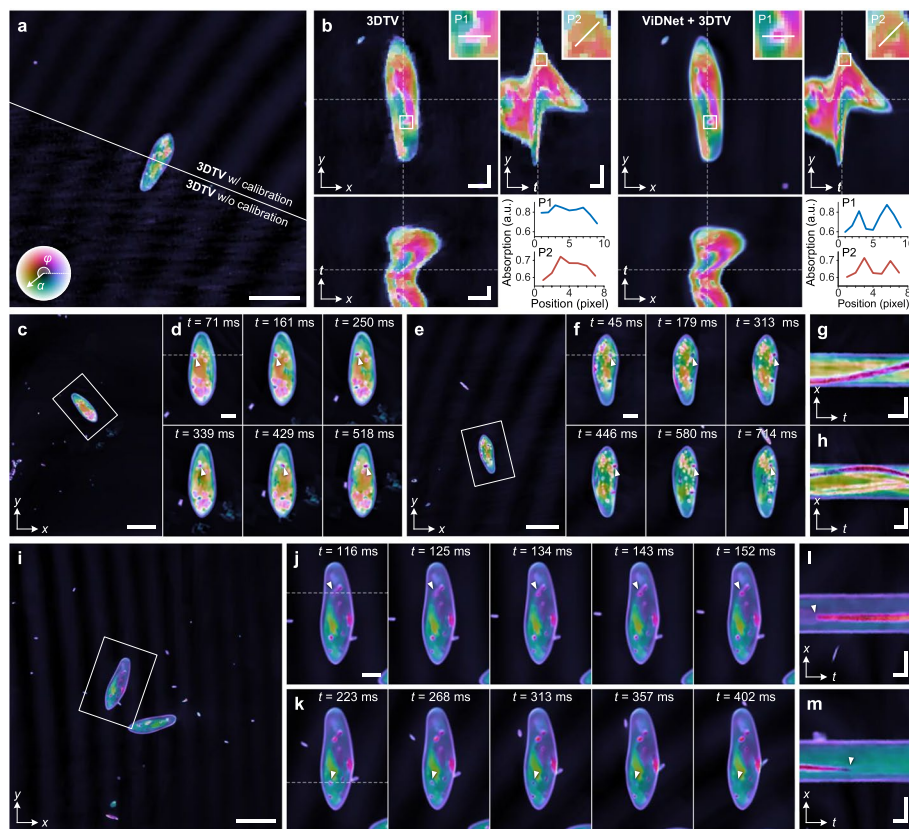


Fig. 3 Holographic imaging of live paramecia. **a** Illustration of the algorithmic self calibration (Supplementary Video 2). **b** The xy , xt and yt cross-sectional views of the spatiotemporal datacube reconstructed from a fast moving paramecium. Enlarged views are displayed in the insets, and the corresponding absorption profiles P1 and P2 are shown (Supplementary Video 3). **c-h** Time-lapse holographic imaging of tracked paramecia exhibiting rotation movements, where the food vacuoles are indicated by the arrows (Supplementary Video 4). **d** and **f** present the enlarged ROIs of **(c)** and **(e)**, respectively. **g** and **h** present the xt cross-sectional views of **(c)** and **(e)**, respectively. **i-m** Time-lapse holographic imaging of a tracked paramecium, where the formation **(j)** and disappearance **(k)** of its food vacuoles are observed (Supplementary Video 5). **l** and **m** present the xt cross-sectional views of **(j)** and **(k)**, respectively. Complex-valued sample transmission profiles are visualized by representing the absorption α via lightness and mapping the wrapped phase φ into a cyclic color map. Scale bars, 200 μm (**a**, **c**, **e**, **i**), 50 μm (xy) and 0.1 s (t) (**b**, **d**, **f-h**, **j-m**)

Discussion

In this work, we have presented a general model-based deep learning framework for dynamic scene recovery in multi-shot computational microscopy. By leveraging spatiotemporal features learned from large-scale video datasets and an optimally designed measurement scheme, we demonstrated physically reliable, time-resolved lensless holographic imaging of dynamic samples, significantly outperforming conventional approaches based on **sparsity or image priors**. The enhanced temporal resolution facilitates the experimental observation of rapid biological activities of paramecia and rotifers, such as food vacuole formation and rupture, which occur on millisecond timescales.

Interestingly, we have proposed that natural intensity video datasets can be successfully applied to holographic video reconstruction of biological samples, addressing hallucination and generalizability challenges common in deep learning microscopy [34]. This is made possible by the optimization of both measurement design and reconstruction,

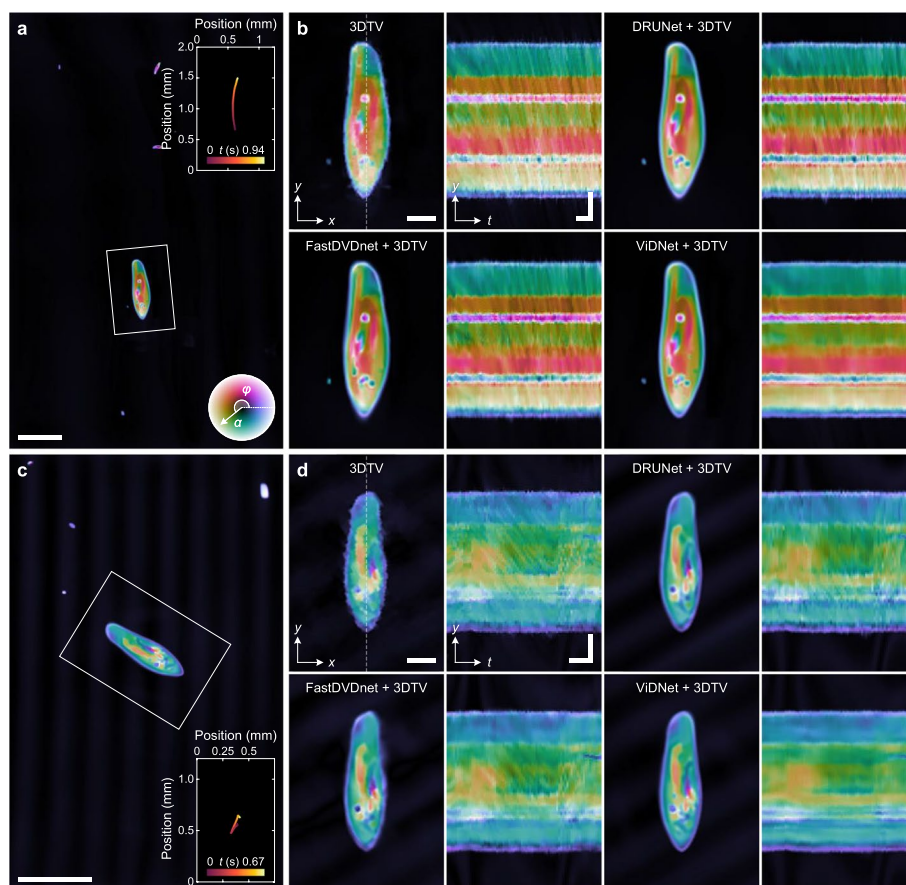


Fig. 4 Comparison of different PnP priors on live parametia. **a, c** Holographic images of fast moving parametium samples. The insets show the sample trajectories. **b, d** Enlarged ROIs and spatiotemporal slices of the tracked parametia corresponding to **(a)** and **(c)**, respectively. Holographic videos of the reconstructed samples can be found in Supplementary Videos 6 and 7, respectively. Scale bars, 200 μm (**a, c**), 50 μm (x, y) and 0.1 s (t) (**b, d**)

distinguishing our work from prior efforts focused solely on algorithmic improvements. Specifically, ViDNet is trained as an AWGN denoiser that captures the low-level spatiotemporal features intrinsic to most dynamic processes, while an information-efficient measurement scheme ensures that strong physical constraints are enforced during the PnP iterations, guiding the network priors toward physically reliable solutions. Despite the promising results achieved through cross-domain knowledge transfer, certain domain-specific priors, such as the spatiotemporal characteristics of diffracted wavefields, are not explicitly embedded within the network, thereby limiting ViDNet's capability to handle defocused or multiple-scattering samples. We believe that the imaging performance could be further improved through fine-tuning the network on domain-specific datasets, when such data are available.

From a technical perspective, several avenues remain for further improvement. The spatiotemporal resolution and the overall data throughput of the current system are primarily limited by the image sensor. Pixel super-resolution techniques, which model the sensor's pixel response to resolve subpixel spatial information up to the diffraction limit [109, 110], and temporal compressive sensing methods, which exploit high-speed

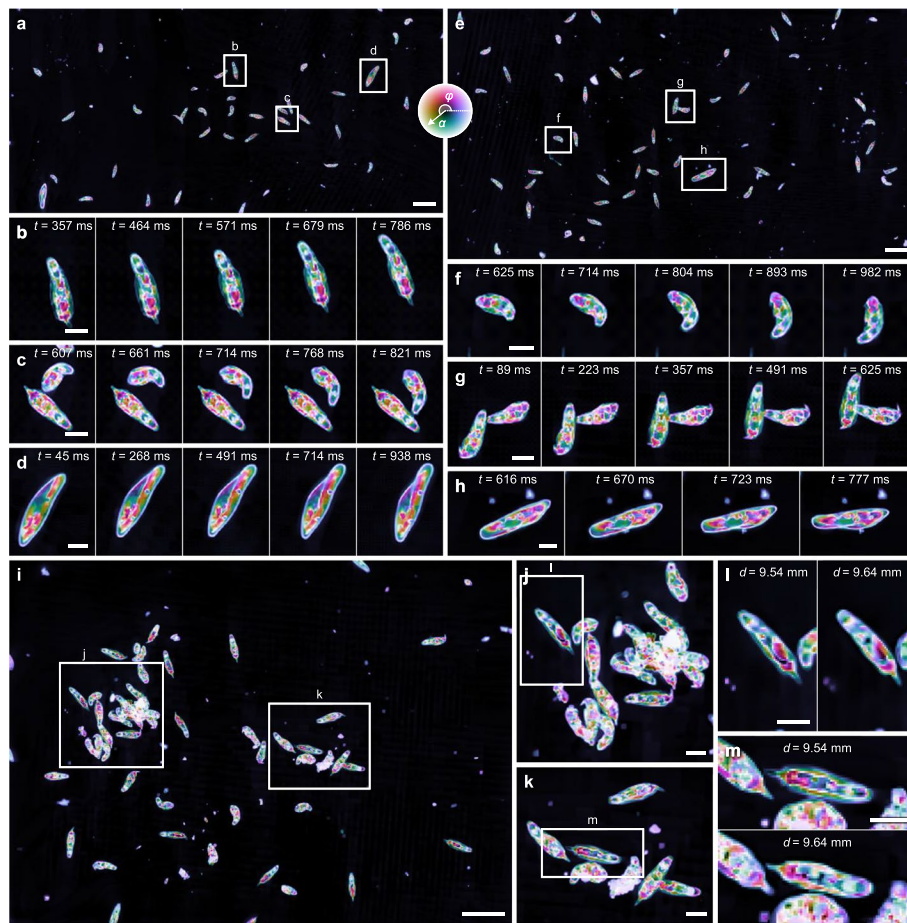


Fig. 5 High-throughput holographic imaging of rotifer and paramecium samples. **a, e** Full-FOV holographic image of the sample (Supplementary Video 8). **b-d** and **f-h** are the time-lapse images of the enlarged ROIs in **(a)** and **(e)**, respectively. **i** Holographic imaging of dense rotifer colonies (Supplementary Video 9). **j, k** Enlarged ROIs in **(i)**. **l** and **m** show the reconstructed fields at different sample-to-sensor distances d for the enlarged ROIs. Scale bars, 200 μm (**a, e, i**), 50 μm (**b-d, f-h, j-m**)

encoding beyond the sensor frame rate [111–113], could be incorporated to further enhance spatial and temporal resolution. The PnP framework can also potentially benefit from advanced network architectures or deep learning models that capture spatiotemporal priors more efficiently [58, 114]. Additionally, while this work develops the measurement scheme and reconstruction algorithm separately, future studies could explore end-to-end design frameworks that jointly optimize hardware and software for improved task-specific imaging performance.

The proposed computational framework could also be readily extended to dynamic scene recovery in other multi-shot computational imaging modalities by adapting to specific forward models. ViDNet learns the low-level spatiotemporal features inherent to natural dynamics, and can be used across various domains, which is particularly beneficial for certain imaging modalities where clean datasets are difficult to acquire. Furthermore, ViDNet could also be incorporated together with other forms of data priors through consensus equilibrium [115], offering additional flexibility for diverse imaging scenarios.

Methods

Experimental setup

As shown in Supplementary Fig. 1, a 532 nm solid-state laser (MGL-III-532, Changchun New Industries Optoelectronics Technology) is used to provide coherent illumination. The laser beam is coupled through a single-mode fiber into the system and collimated by a plano-convex lens (focal length 100 mm). The samples are placed on a sample holder, which is translated at a speed of ~ 2 mm/s by a motorized linear translation stage (MMTP65-25, MFOPT) during the measurement. The diffuser is designed to provide random binary amplitude modulation to the wavefront and is fabricated by patterning a chrome coating on a glass substrate via laser direct writing. It has a feature size of $16\ \mu\text{m}$ and a total dimension large enough to cover the entire sensor FOV. The diffuser is also mounted on a 2D motorized stage (KMTS25E/M, Thorlabs) as required by the calibration experiment. The intensity images are captured with an industrial board-level complementary metal-oxide semiconductor image sensor (Alvium 1800 U-811 m, Allied Vision). The configuration is optimized to maintain a compact form, with ~ 7 mm sample-to-diffuser and ~ 3 mm diffuser-to-sensor distances. For the biological experiments, the paramecium and rotifer samples can be found in natural environment or acquired from a biological supply company. A small drop of water containing these organisms is sandwiched between two coverslips (CG00K1, Thorlabs) and placed on the sample holder for observation. It is worth noting that in coded ptychography, measurement diversity can be achieved as long as there are lateral displacements between the sample and the diffuser [88]. Therefore, different hardware implementations have been adopted in prior works, either translating the sample or the diffuser. In this work, we adopt the former approach, as translating the sample enables the decoupling of both the diffuser and probe profiles from the sample's transmission function, allowing high-quality sample field reconstruction even under imperfect diffuser or probe calibration.

System calibration

According to the forward model of Eq. (1), the imaging system involves several key parameters that should be calibrated to accurately model the imaging process. The entire calibration and data processing pipeline is summarized in Supplementary Note 5. The sample-to-sensor distance ($d_1 + d_2$) and the diffuser to sensor distance (d_2) were determined by digital refocusing using an additional diffraction image captured with only the sample or diffuser placed in the light path. The diffuser modulation profile $\mathbf{m}$ was calibrated in advance by conventional near-field ptychography, where the diffuser was illuminated by a confined probe and laterally translated to overlapping scanning positions during the measurement. A modified extended ptychographic iterative engine was used to retrieve the diffuser profile from the ptychographic dataset. To estimate the translation positions $\mathbf{r}_k$ at each time point, we first performed frame-by-frame reconstruction using only the spatial-domain 2DTV regularizer. The intermediate results, albeit still corrupted by artifacts, can guide image registration using properly selected control points [116].

Despite relatively precise system calibration, reconstruction artifacts from measurement imperfections remain visible in the sample field (Fig. 3a). To further compensate for

model mismatch, we introduced an algorithmic self-calibration scheme to refine the system parameters by solving the following multi-parameter optimization problem [117]:

$$\hat{\mathbf{x}}, \hat{\mathbf{p}} = \underset{\mathbf{x}, \mathbf{p}}{\operatorname{argmin}} \left\{ \frac{1}{2} \sum_{k=1}^K \left\| \mathbf{F} \operatorname{diag}(\mathbf{p}) \mathbf{T}(\mathbf{r}_k) \mathbf{x}_k - \mathbf{y}_k \right\|_2^2 + 3\text{DTV}(\mathbf{x}) \right\}, \quad (7)$$

where $\mathbf{F} = \mathbf{Q}(d_2) \operatorname{diag}(\mathbf{m}) \mathbf{Q}(d_1)$ denotes the forward measurement operator. The above formulation is similar to the reconstruction of both the probe and the sample field in classical ptychography [118, 119]. Notably, although only the probe function $\mathbf{p}$ serves as the system's learnable parameter, it may also account for other contributing factors including the inaccuracy of the diffuser calibration, multiple reflections and other experimental imperfections [120]. An alternating minimization approach is adopted to solve Eq. (7), as detailed in Supplementary Note 6.

Network architecture and training details

The architecture of the proposed ViDNet is illustrated in Supplementary Fig. 3a. To fully exploit the inter-frame temporal priors of dynamic scenes, ViDNet takes five consecutive frames together with the noise level as input, and outputs the denoised central frame. It adopts a two-stage coarse-to-fine architecture, which has been proven to be effective for video processing [101, 121]. In the first stage, every three neighboring frames within the sequence are processed separately by a first denoising block. In the second stage, the intermediate results are processed by a second denoising block for further refinement, and the denoised frame is obtained. The two denoising blocks have different learnable weights but share the same architecture, which is a modified U-Net model with four down-sampling and up-sampling modules, as shown in Supplementary Fig. 3b. Each module consists of four successive residual blocks with two 3×3 convolution layers and a rectified linear unit activation function. Notably, ViDNet does not incorporate any bias parameters. All bias terms in the convolutional layers are disabled, and bias-inducing operations such as batch normalization are also excluded. Such a design could enhance the scalability of the network and improves its numerical stability when applied to PnP optimization (Supplementary Note 4) [108, 122]. It is worth noticing that even though the network itself only captures spatiotemporal features within five neighboring frames, longer-range spatiotemporal consistency can be realized because of the iterative nature of PnP optimization.

The training dataset consists of 70,405 five-frame grayscale video clips of natural dynamic scenes extracted from the DAVIS [123] and Vimeo-90K database [124]. During training, 16 patches with dimensions $128 \times 128 \times 5$ are randomly sampled from the dataset and added with AWGN, with noise standard deviation (SD) uniformly selected within the range of 0 to 50. Data augmentation techniques including random crop, flip, rotation and time reversal are applied to the input videos to enhance dataset diversity. The network parameters are optimized by minimizing the ℓ_1 loss between the network output and the ground truth using the Adam optimizer, which has been shown to provide better robustness against outliers and more stable convergence during the training stage [108]. The learning rate follows a scheduled decay, initialized at 1×10^{-4} and halved every 50 epochs until reaching the termination threshold of 1×10^{-7} .

Orthogonalization is applied to the convolution kernels repeatedly during the training process to boost the learning efficiency [125]. The entire training process takes approximately 10 days on an NVIDIA A100 graphics processing unit (GPU) card.

Algorithm implementation

As conceptually shown in Fig. 1c and detailed in Supplementary Fig. 4, PnP-FISTA is built upon FISTA with the proximal update replaced by a blackbox AWGN denoising step. To process complex-valued videos with a real-valued network, we perform denoising to the real and imaginary parts of the input video separately [126]. The main advantages of this approach over other processing schemes (e.g., separating complex-valued images into amplitude and phase channels) are twofold. First, the operation of extracting the real and imaginary parts is linear with respect to the complex amplitude and is thus consistent with the PnP optimization theory (Supplementary Note 2). Second, the spatial textures of the real and imaginary parts bear much resemblance to natural scenes that are used for training, whereas phase images often have spatial discontinuities caused by phase wraps (Supplementary Fig. 5). In the experiments, we observed that purely network-based denoisers tend to amplify reconstruction artifacts due to experimental imperfections and measurement noise. Therefore, an additional 3DTV-based refinement step is introduced after network denoising for more stable and robust recovery [127], leading to the following steps:

$$\mathbf{w}_r^{(i)} = \mathcal{D}_{\sigma^{(i)}}(\mathbf{u}_r^{(i)}) \quad (8)$$

$$\mathbf{w}_i^{(i)} = \mathcal{D}_{\sigma^{(i)}}(\mathbf{u}_i^{(i)}) \quad (9)$$

$$\mathbf{w}^{(i)} = \mathbf{w}_r^{(i)} + j\mathbf{w}_i^{(i)} \quad (10)$$

$$\mathbf{v}^{(i)} = \text{prox}_{\gamma\text{3DTV}}(\mathbf{w}^{(i)}), \quad (11)$$

where $\mathbf{u}_r^{(i)} = \text{Re}(\mathbf{u}^{(i)})$, $\mathbf{u}_i^{(i)} = \text{Im}(\mathbf{u}^{(i)})$, and j denotes the imaginary unit. The denoising strength $\sigma^{(i)}$ was set to exponentially decrease as the algorithm proceeds (Supplementary Note 7). Intuitively, a larger denoising strength helps accelerate convergence at the beginning stage, while the final value balances between the model and data priors as the algorithm converges. The 3DTV regularizer can be expressed as

$$\text{3DTV}(\mathbf{x}) = \rho_s \|\mathbf{D}_x \mathbf{x}\|_1 + \rho_s \|\mathbf{D}_y \mathbf{x}\|_1 + \rho_t \|\mathbf{D}_t \mathbf{x}\|_1, \quad (12)$$

where $\mathbf{D}_x$, $\mathbf{D}_y$ and $\mathbf{D}_t$ denote the gradient operators along the x , y and t axes, respectively. ρ_s and ρ_t are the regularization parameters applied to the spatial and temporal domains, respectively. The proximal update of Eq. (11) can be readily solved via iterative algorithms [57]. When denoising video sequences with ViDNet, a sliding window of five frames is used with a step size of one and symmetric boundary condition. As a result, the reconstruction quality of the first and last two frames in the sequence is slightly compromised, as can be observed from the reduced inter-frame correlations in Fig. 2b.

In all experiments, we first performed 200 iterations of 3DTV-regularized reconstruction [57] to obtain a computationally efficient, albeit coarse, initial estimate. This intermediate result served as the initialization for our algorithmic self-calibration procedure, which jointly refined both the sample and probe profiles through 50 additional iterations. Then, the estimated sample field, together with the updated forward model, was fed into the deep PnP-FISTA framework for learning-based PnP reconstruction. Thanks to the warm start and acceleration techniques, typically 20–30 iterations are sufficient for the PnP-ViDNet+3DTV algorithm to provide competitive results, which takes ~ 9 minutes on an A100 GPU to reconstruct a $1,264 \times 1,028 \times 20$ holographic video. Finally, we applied aberration correction through polynomial fitting to compensate for background phase variations in the reconstructed sample field.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s43074-025-00222-2>.

Supplementary Material 1

Authors' contributions

Y.G. and L.C. conceived the idea and initiated the project. Y.G. performed the experiments and data analysis. Y.G. and L.C. wrote the manuscript. L.C. acquired funding and supervised the project.

Funding

This work was supported by the National Natural Science Foundation of China (62235009, 624B2080).

Data availability

All experimental data reported in this work are available via Zenodo at <https://doi.org/10.5281/zenodo.17523561>. The code for video PnP algorithms is available at <https://github.com/THUHoloLab/STRIVER-deep>. The training code for ViDNet and the pretrained models are available at <https://github.com/THUHoloLab/ViDNet>.

Declarations

Competing interests

The authors declare no competing interests.

Received: 25 July 2025 Revised: 10 November 2025 Accepted: 17 December 2025

Published online: 07 January 2026

References

- Zheng G, Horstmeyer R, Yang C. Wide-field, high-resolution Fourier ptychographic microscopy. *Nat Photon.* 2013;7(9):739–45. <https://doi.org/10.1038/nphoton.2013.187>.
- Heintzmann R, Huser T. Super-resolution structured illumination microscopy. *Chem Rev.* 2017;117(23):13890–908. <https://doi.org/10.1021/acs.chemrev.7b00218>.
- Pfeiffer F. X-ray ptychography. *Nat Photon.* 2018;12(1):9–17. <https://doi.org/10.1038/s41566-017-0072-5>.
- Miao J. Computational microscopy with coherent diffractive imaging and ptychography. *Nature.* 2025;637(8045):281–95. <https://doi.org/10.1038/s41586-024-08278-z>.
- Park Y, Depeursinge C, Popescu G. Quantitative phase imaging in biomedicine. *Nat Photon.* 2018;12(10):578–89. <https://doi.org/10.1038/s41566-018-0253-x>.
- Shaked NT, Boppart SA, Wang LV, Popp J. Label-free biomedical optical imaging. *Nat Photon.* 2023;17(12):1031–41. <https://doi.org/10.1038/s41566-023-01299-6>.
- Huang Z, Cao L. Quantitative phase imaging based on holography: trends and new perspectives. *Light Sci Appl.* 2024;13(1):145. <https://doi.org/10.1038/s41377-024-01453-x>.
- Ma H, Chen M, Xu J, Yang Y, Zhao Y, Liu Y. An omni-mesoscope for multiscale high-throughput quantitative phase imaging of cellular dynamics and high-content molecular characterization. *Sci Adv.* 2024;10(42):5009. <https://doi.org/10.1126/sciadv.adq5009>.
- Katz O, Heidmann P, Fink M, Gigan S. Non-invasive single-shot imaging through scattering layers and around corners via speckle correlations. *Nat Photon.* 2014;8(10):784–90. <https://doi.org/10.1038/nphoton.2014.189>.
- Yoon S, Kim M, Jang M, Choi Y, Choi W, Kang S, et al. Deep optical imaging within complex scattering media. *Nat Rev Phys.* 2020;2(3):141–58. <https://doi.org/10.1038/s42254-019-0143-2>.
- Haim O, Boger-Lombard J, Katz O. Image-guided computational holographic wavefront shaping. *Nat Photon.* 2025;19(1):44–53. <https://doi.org/10.1038/s41566-024-01544-6>.

12. Wu J, Lu Z, Jiang D, Guo Y, Qiao H, Zhang Y, Zhu T, Cai Y, Zhang X, Zhanghao K, et al. Iterative tomography with digital adaptive optics permits hour-long intravital observation of 3D subcellular dynamics at millisecond scale. *Cell*. 2021;184(12):3318–32. <https://doi.org/10.1016/j.cell.2021.04.029>.
13. Kim G, Hugonnet H, Kim K, Lee J-H, Lee SS, Ha J, et al. Holotomography. *Nat Rev Methods Primers*. 2024;4(1):51. <https://doi.org/10.1038/s43586-024-00327-1>.
14. Hsieh H-C, Han Q, Brenes D, Bishop KW, Wang R, Wang Y, et al. Imaging 3D cell cultures with optical microscopy. *Nat Methods*. 2025. <https://doi.org/10.1038/s41592-025-02647-w>.
15. Gao L, Wang LV. A review of snapshot multidimensional optical imaging: measuring photon tags in parallel. *Phys Rep*. 2016;616:1–37. <https://doi.org/10.1016/j.physrep.2015.12.004>.
16. Fan J, Suo J, Wu J, Xie H, Shen Y, Chen F, et al. Video-rate imaging of biological dynamics at centimetre scale and micrometre resolution. *Nat Photon*. 2019;13(11):809–16. <https://doi.org/10.1038/s41566-019-0474-7>.
17. Chan AC, Kim J, Pan A, Xu H, Nojima D, Hale C, et al. Parallel Fourier ptychographic microscopy for high-throughput screening with 96 cameras (96 Eyes). *Sci Rep*. 2019;9(1):11114. <https://doi.org/10.1038/s41598-019-47146-z>.
18. Zhou KC, Harfouche M, Cooke CL, Park J, Konda PC, Kreiss L, et al. Parallelized computational 3D video microscopy of freely moving organisms at multiple gigapixels per second. *Nat Photon*. 2023;17(5):442–50. <https://doi.org/10.1038/s41566-023-01171-7>.
19. Chen N, Brady D. Ptychographic wavefront cameras. *Opt Lett*. 2024;49(23):6653–6. <https://doi.org/10.1364/OL.542871>.
20. Zhou KC, Cook C, Chakraborty A, Bagwell J, Jönsson J, Lee KC, et al. High-speed 4D fluorescence light field tomography of whole freely moving organisms. *Optica*. 2025;12(5):674–84. <https://doi.org/10.1364/OPTICA.549707>.
21. Kumar A, McNally KE, Zhang Y, Haslett-Saunders A, Wang X, Guillem-Marti J, et al. Multispectral live-cell imaging with uncompromised spatiotemporal resolution. *Nat Photon*. 2025. <https://doi.org/10.1038/s41566-025-01745-7>.
22. Tian L, Liu Z, Yeh L-H, Chen M, Zhong J, Waller L. Computational illumination for high-speed in vitro Fourier ptychographic microscopy. *Optica*. 2015;2(10):904–11. <https://doi.org/10.1364/OPTICA.2.000904>.
23. Sidorenko P, Cohen O. Single-shot ptychography. *Optica*. 2015;3(1):9–14. <https://doi.org/10.1364/OPTICA.3.000009>.
24. Sun J, Chen Q, Zhang J, Fan Y, Zuo C. Single-shot quantitative phase microscopy based on color-multiplexed fourier ptychography. *Opt Lett*. 2018;43(14):3365–8. <https://doi.org/10.1364/OL.43.003365>.
25. Li J, Matlock A, Li Y, Chen Q, Zuo C, Tian L. High-speed in vitro intensity diffraction tomography. *Adv Photon*. 2019;1(6):066004. <https://doi.org/10.1117/1.AP.1.6.066004>.
26. Kwon H, Arbabi E, Kamali SM, Faraji-Dana M, Faraon A. Single-shot quantitative phase gradient microscopy using a system of multifunctional metasurfaces. *Nat Photon*. 2020;14(2):109–14. <https://doi.org/10.1038/s41566-019-0536-x>.
27. Yanny K, Antipa N, Liberti W, Dehaeck S, Monakhova K, Liu FL, et al. Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy. *Light Sci Appl*. 2020;9(1):171. <https://doi.org/10.1038/s41377-020-00403-7>.
28. Ge B, He Y, Deng M, Habibur Rahman M, Wang Y, Wu Z, Wong CHN, Chan MK, Ho Y-P, Duan L, et al. Single-frame label-free cell tomography at speed of more than 10,000 volumes per second. 2202. [arXiv:2202.03627](https://arxiv.org/abs/2202.03627).
29. Li L, Wang S, Zhao F, Zhang Y, Wen S, Chai H, et al. Single-shot deterministic complex amplitude imaging with a single-layer metalens. *Sci Adv*. 2024;10(1):0501. <https://doi.org/10.1126/sciadv.adl0501>.
30. Zheng C, Zhang S, Liu S, Yang D, Hao Q. Single-shot Fourier ptychographic microscopy with isotropic lateral resolution via polarization-multiplexed led illumination. *Biomed Opt Express*. 2024;15(2):672–86. <https://doi.org/10.1364/BOE.513684>.
31. Zhang Y, Bai L, Wang X, Zhao Y, Zhang T, Ye L, et al. Super-resolution imaging of fast morphological dynamics of neurons in behaving animals. *Nat Methods*. 2025;22(1):177–86. <https://doi.org/10.1038/s41592-024-02535-9>.
32. Yoon H, Chae H, Lee KC, Lee K, Lee SA. Single-shot Fourier ptychography using polarization-encoded illumination. *Appl Phys Lett*. 2025;126(12):123702. <https://doi.org/10.1063/5.0248101>.
33. Barbastathis G, Ozcan A, Situ G. On the use of deep learning for computational imaging. *Optica*. 2019;6(8):921–43. <https://doi.org/10.1364/OPTICA.6.000921>.
34. Belthangady C, Royer LA. Applications, promises, and pitfalls of deep learning for fluorescence image reconstruction. *Nat Methods*. 2019;16(12):1215–25. <https://doi.org/10.1038/s41592-019-0458-z>.
35. Park J, Bai B, Ryu D, Liu T, Lee C, Luo Y, et al. Artificial intelligence-enabled quantitative phase imaging methods for life sciences. *Nat Methods*. 2023;20(11):1645–60. <https://doi.org/10.1038/s41592-023-02041-4>.
36. Wang K, Song L, Wang C, Ren Z, Zhao G, Dou J, et al. On the use of deep learning for phase recovery. *Light Sci Appl*. 2024;13(1):4. <https://doi.org/10.1038/s41377-023-01340-x>.
37. Zeng T, Zhu Y, Lam EY. Deep learning for digital holography: a review. *Opt Express*. 2021;29(24):40572–93. <https://doi.org/10.1364/OE.443367>.
38. Luo X, Wang S, Liu J, Dong X, He P, Yang Q, et al. Revolutionizing optical imaging: computational imaging via deep learning. *Photonics Insights*. 2025;4(2):03. <https://doi.org/10.3788/PI.2025.R03>.
39. Rivenson Y, Göröcs Z, Günaydin H, Zhang Y, Wang H, Ozcan A. Deep learning microscopy. *Optica*. 2017;4(11):1437–43. <https://doi.org/10.1364/OPTICA.4.001437>.
40. Rivenson Y, Zhang Y, Günaydin H, Teng D, Ozcan A. Phase recovery and holographic image reconstruction using deep learning in neural networks. *Light Sci Appl*. 2018;7(2):17141–17141.
41. Nehme E, Weiss LE, Michaeli T, Shechtman Y. Deep-STORM: super-resolution single-molecule microscopy by deep learning. *Optica*. 2018;5(4):458–64. <https://doi.org/10.1364/OPTICA.5.000458>.
42. Wang F, Bian Y, Wang H, Lyu M, Pedrini G, Osten W, et al. Phase imaging with an untrained neural network. *Light Sci Appl*. 2020;9(1):77. <https://doi.org/10.1038/s41377-020-0302-3>.
43. Qiao C, Li D, Guo Y, Liu C, Jiang T, Dai Q, et al. Evaluation and development of deep neural networks for image super-resolution in optical microscopy. *Nat Methods*. 2021;18(2):194–202. <https://doi.org/10.1038/s41592-020-01048-5>.

44. Seong B, Kim W, Kim Y, Hyun K-A, Jung H-I, Lee J-S, et al. E2E-BPF microscope: extended depth-of-field microscopy using learning-based implementation of binary phase filter and image deconvolution. *Light Sci Appl*. 2023;12(1):269. <https://doi.org/10.1038/s41377-023-01300-5>.
45. Saguy A, Alalouf O, Opatovski N, Jang S, Heilemann M, Shechtman Y. DBlink: dynamic localization microscopy in super spatiotemporal resolution via deep learning. *Nat Methods*. 2023;20(12):1939–48. <https://doi.org/10.1038/s41592-023-01966-0>.
46. Lee C, Song G, Kim H, Ye JC, Jang M. Deep learning based on parameterized physical forward model for adaptive holographic imaging with unpaired data. *Nat Mach Intell*. 2023;5(1):35–45. <https://doi.org/10.1038/s42256-022-00584-3>.
47. Huang L, Chen H, Liu T, Ozcan A. Self-supervised learning of hologram reconstruction using physics consistency. *Nat Mach Intell*. 2023;5(8):895–907. <https://doi.org/10.1038/s42256-023-00704-7>.
48. Zhang O, Zhou H, Feng BY, Larsson EM, Alcalde RE, Yin S, et al. Single-shot volumetric fluorescence imaging with neural fields. *Adv Photon*. 2025;7(2):026001. <https://doi.org/10.1117/1.AP.7.2.026001>.
49. Qian J, Wang C, Wu H, Chen Q, Zuo C. Ensemble deep learning-enabled single-shot composite structured illumination microscopy (edl-csim). *Photonix*. 2025;6(1):13. <https://doi.org/10.1186/s43074-025-00171-w>.
50. Zhang S, Pan A, Sun H, Tan Y, Cao L. High-fidelity computational microscopy via feature-domain phase retrieval. *Adv Sci*. 2025;12:2413975. <https://doi.org/10.1002/advs.202413975>.
51. Cover TM, Thomas JA. *Elements of Information Theory*. Hoboken, New Jersey: John Wiley & Sons; 2012.
52. Huang X, Fan J, Li L, Liu H, Wu R, Wu Y, et al. Fast, long-term, super-resolution imaging with Hessian structured illumination microscopy. *Nat Biotechnol*. 2018;36(5):451–9. <https://doi.org/10.1038/nbt.4115>.
53. Yoo J, Jin KH, Gupta H, Yerly J, Stuber M, Unser M. Time-dependent deep image prior for dynamic MRI. *IEEE Trans Med Imaging*. 2021;40(12):3337–48. <https://doi.org/10.1109/TMI.2021.3084288>.
54. Li X, Li Y, Zhou Y, Wu J, Zhao Z, Fan J, et al. Real-time denoising enables high-sensitivity fluorescence time-lapse imaging beyond the shot-noise limit. *Nat Biotechnol*. 2023;41(2):282–92. <https://doi.org/10.1038/s41587-022-01450-8>.
55. Li X, Hu X, Chen X, Fan J, Zhao Z, Wu J, et al. Spatial redundancy transformer for self-supervised fluorescence image denoising. *Nat Comput Sci*. 2023;3(12):1067–80. <https://doi.org/10.1038/s43588-023-00568-2>.
56. Feng BY, Guo H, Xie M, Boominathan V, Sharma MK, Veeraraghavan A, et al. NeuWS: neural wavefront shaping for guidestar-free imaging through static and dynamic scattering media. *Sci Adv*. 2023;9(26):4671. <https://doi.org/10.1126/sciadv.adg4671>.
57. Gao Y, Cao L. Motion-resolved, reference-free holographic imaging via spatiotemporally regularized inversion. *Optica*. 2024;11(1):32–41. <https://doi.org/10.1364/OPTICA.506572>.
58. Liang J, Cao J, Fan Y, Zhang K, Ranjan R, Li Y, et al. VRT: a video restoration transformer. *IEEE Trans Image Process*. 2024. <https://doi.org/10.1109/TIP.2024.3372454>.
59. Cao R, Divekar NS, Nuñez JK, Upadhyayula S, Waller L. Neural space–time model for dynamic multi-shot imaging. *Nat Methods*. 2024;21(12):2336–41. <https://doi.org/10.1038/s41592-024-02417-0>.
60. Qiao C, Liu S, Wang Y, Xu W, Geng X, Jiang T, Zhang J, Meng Q, Qiao H, Li D, et al. A neural network for long-term super-resolution imaging of live cells with reliable confidence quantification. *Nat Biotechnol*. 2025. <https://doi.org/10.1038/s41587-025-02553-8>.
61. Lee W, Jang MA, Nam HS, Song J, Choi J, Song JW, et al. Self-supervised denoising of dynamic fluorescence images via temporal gradient-empowered deep learning. *Photonix*. 2025;6(1):15. <https://doi.org/10.1186/s43074-025-00173-8>.
62. Kwon T, Song G, Kim Y, Kim J, Ye JC, Jang M. Video diffusion posterior sampling for seeing beyond dynamic scattering layers. *IEEE Trans Pattern Anal Mach Intell*. 2025. <https://doi.org/10.1109/TPAMI.2025.3598457>.
63. Zhang B, Wu Z, Feng BT, Song Y, Yue Y, Bouman KL. STeP: A general and scalable framework for solving video inverse problems with spatiotemporal diffusion priors. 2025. arXiv preprint [arXiv:2504.07549](https://arxiv.org/abs/2504.07549).
64. Greenbaum A, Luo W, Su T-W, Göröcs Z, Xue L, Isikman SO, et al. Imaging without lenses: achievements and remaining challenges of wide-field on-chip microscopy. *Nat Methods*. 2012;9(9):889–95. <https://doi.org/10.1038/nmeth.2114>.
65. Lee SA, Yang C. A smartphone-based chip-scale microscope using ambient illumination. *Lab Chip*. 2014;14(16):3056–63. <https://doi.org/10.1039/C4LC00523F>.
66. Potter CJ, Xiong Z, McLeod E. Clinical and biomedical applications of lensless holographic microscopy. *Laser Photon Rev*. 2024;18(10):2400197. <https://doi.org/10.1002/lpor.202400197>.
67. Fan Y, Li J, Lu L, Sun J, Hu Y, Zhang J, Li Z, Shen Q, Wang B, Zhang R, et al. Smart computational light microscopes (SCLMs) of smart computational imaging laboratory (SCILab). *Photonix*. 2021;2:1–64. <https://doi.org/10.1186/s43074-021-00040-2>.
68. You J, Lee D, Song G, Lee C, Jang M. Motion-free high-resolution on-chip microscopy using led matrix. *Opt Express*. 2024;32(21):36549–62. <https://doi.org/10.1364/OE.537127>.
69. Rivenson Y, Liu T, Wei Z, Zhang Y, De Haan K, Ozcan A. PhaseStain: the digital staining of label-free quantitative phase microscopy images using deep learning. *Light Sci Appl*. 2019;8(1):23. <https://doi.org/10.1038/s41377-019-0129-y>.
70. Jiang S, Guo C, Song P, Zhou N, Bian Z, Zhu J, et al. Resolution-enhanced parallel coded ptychography for high-throughput optical imaging. *ACS Photonics*. 2021;8(11):3261–71. <https://doi.org/10.1021/acsp Photonics.1c01085>.
71. Sun J, Wu J, Wu S, Goswami R, Girardo S, Cao L, et al. Quantitative phase imaging through an ultra-thin lensless fiber endoscope. *Light Sci Appl*. 2022;11(1):204. <https://doi.org/10.1038/s41377-022-00898-2>.
72. Sun J, Kuschmierz R, Katz O, Koukourakis N, Czarnecki JW. Lensless fiber endomicroscopy in biomedicine. *Photonix*. 2024;5(1):18. <https://doi.org/10.1186/s43074-024-00133-8>.
73. Song P, Wang R, Loetgering L, Liu J, Vouras P, Lee Y, et al. Ptycho-endoscopy on a lensless ultrathin fiber bundle tip. *Light Sci Appl*. 2024;13(1):168. <https://doi.org/10.1038/s41377-024-01510-5>.
74. Berdeu A, Laperrousaz B, Bordy T, Mandula O, Morales S, Gidrol X, et al. Lens-free microscopy for 3D+ time acquisitions of 3D cell culture. *Sci Rep*. 2018;8(1):16135. <https://doi.org/10.1038/s41598-018-34253-6>.

75. Mirecki B, Rogalski M, Arcab P, Rogujski P, Stanaszek L, Józwiak M, et al. Low-intensity illumination for lensless digital holographic microscopy with minimized sample interaction. *Biomed Opt Express*. 2022;13(11):5667–82. <https://doi.org/10.1364/BOE.464367>.
76. Arcab P, Rogalski M, Marzejon M, Rogujski P, Stanaszek L, Trusiak M. Multi-culture label-free quantitative cell migration sensing with single-cell precision. *Biomed Opt Express*. 2024;16(1):222–34. <https://doi.org/10.1364/BOE.541010>.
77. Kim J, Song S, Kim H, Kim B, Park M, Oh SJ, et al. Ptychographic lens-less birefringence microscopy using a mask-modulated polarization image sensor. *Sci Rep*. 2023;13(1):19263. <https://doi.org/10.1038/s41598-023-46496-z>.
78. Yang L, Wang R, Zhao Q, Song P, Jiang S, Wang T, et al. Lensless polarimetric coded ptychography for high-resolution, high-throughput gigapixel birefringence imaging on a chip. *Photonics Res*. 2023;11(12):2242–55. <https://doi.org/10.1364/PRJ.504378>.
79. Wu Y-C, Shiledar A, Li Y-C, Wong J, Feng S, Chen X, et al. Air quality monitoring using mobile microscopy and machine learning. *Light Sci Appl*. 2017;6(9):17046. <https://doi.org/10.1038/lsa.2017.46>.
80. Moon S, Keles HO, Ozcan A, Khademhosseini A, Hæggstrom E, Kuritzkes D, et al. Integrating microfluidics and lensless imaging for point-of-care testing. *Biosens Bioelectron*. 2009;24(11):3208–14. <https://doi.org/10.1016/j.bios.2009.03.037>.
81. Seo S, Su T-W, Tseng DK, Erlinger A, Ozcan A. Lensfree holographic imaging for on-chip cytometry and diagnostics. *Lab Chip*. 2009;9(6):777–87. <https://doi.org/10.1039/B813943A>.
82. Gurkan UA, Moon S, Geckil H, Xu F, Wang S, Lu TJ, et al. Miniaturized lensless imaging systems for cell and micro-organism visualization in point-of-care testing. *Biotechnol J*. 2011;6(2):138–49. <https://doi.org/10.1002/biot.201000427>.
83. Allier C, Morel S, Vincent R, Ghenim L, Navarro F, Menneteau M, et al. Imaging of dense cell cultures by multiwavelength lens-free video microscopy. *Cytometry A*. 2017;91(5):433–42. <https://doi.org/10.1002/cyto.a.23079>.
84. Wu X, Zhou N, Chen Y, Sun J, Lu L, Chen Q, et al. Lens-free on-chip 3D microscopy based on wavelength-scanning Fourier ptychographic diffraction tomography. *Light Sci Appl*. 2024;13(1):237. <https://doi.org/10.1038/s41377-024-01568-1>.
85. Guo C, Jiang S, Song P, Wang T, Shao X, Zhang Z, et al. Quantitative multi-height phase retrieval via a coded image sensor. *Biomed Opt Express*. 2021;12(11):7173–84. <https://doi.org/10.1364/BOE.443528>.
86. Zhou Y, Song W, Wu L, Fan L, Wang J, Jiang S, et al. Large-scale lensless microscopy with fast acquisition and region-wise focusing. *Laser Photon Rev*. 2024;18(10):2400498. <https://doi.org/10.1002/lpor.202400498>.
87. Jiang S, Zhu J, Song P, Guo C, Bian Z, Wang R, et al. Wide-field, high-resolution lensless on-chip microscopy via near-field blind ptychographic modulation. *Lab Chip*. 2020;20(6):1058–65. <https://doi.org/10.1039/C9LC01027K>.
88. Jiang S, Song P, Wang T, Yang L, Wang R, Guo C, et al. Spatial-and Fourier-domain ptychography for high-throughput bio-imaging. *Nat Protoc*. 2023;18(7):2051–83. <https://doi.org/10.1038/s41596-023-00829-4>.
89. Zhang J, Sun J, Chen Q, Zuo C. Resolution analysis in a lens-free on-chip digital holographic microscope. *IEEE Trans Comput Imaging*. 2020;6:697–710. <https://doi.org/10.1109/TCI.2020.2964247>.
90. Lo YH, Zhao L, Gallagher-Jones M, Rana A, Lodico J, Xiao W, et al. In situ coherent diffractive imaging. *Nat Commun*. 2018;9(1):1826. <https://doi.org/10.1038/s41467-018-04259-9>.
91. Zhang J, Yang D, Tao Y, Zhu Y, Lv W, Miao D, et al. Spatiotemporal coherent modulation imaging for dynamic quantitative phase and amplitude microscopy. *Opt Express*. 2021;29(23):38451–64. <https://doi.org/10.1364/OE.434957>.
92. Sheng P, Zhang F. Serial coherent diffraction imaging of dynamic samples based on inter-frame continuity. *Light Sci Appl*. 2025;14:230. <https://doi.org/10.1038/s41377-025-01860-8>.
93. Vaswani N, Nayer S, Eldar YC. Low-rank phase retrieval. *IEEE Trans Signal Process*. 2017;65(15):4059–74. <https://doi.org/10.1109/TSP.2017.2684758>.
94. Sun M, Wang K, Mishra YN, Qiu S, Heidrich W. Space-time Fourier ptychography for in vivo quantitative phase imaging. *Optica*. 2024;11(9):1250–60. <https://doi.org/10.1364/OPTICA.531646>.
95. Kim J, Chowdhury S. Space-time inverse-scattering of translation-based motion. *Optica*. 2025;12(5):643–53. <https://doi.org/10.1364/OPTICA.554264>.
96. Jiang S, Guo C, Bian Z, Wang R, Zhu J, Song P, et al. Ptychographic sensor for large-scale lensless microbial monitoring with high spatiotemporal resolution. *Biosens Bioelectron*. 2022;196:113699. <https://doi.org/10.1016/j.bios.2021.113699>.
97. Venkatakrisnan SV, Bouman CA, Wohlberg B. Plug-and-play priors for model based reconstruction. In: 2013 IEEE Global Conference on Signal and Information Processing. IEEE; 2013. pp. 945–948. <https://doi.org/10.1109/GlobaISIP.2013.6737048>.
98. Kamilov US, Bouman CA, Buzzard GT, Wohlberg B. Plug-and-play methods for integrating physical and learned models in computational imaging: theory, algorithms, and applications. *IEEE Signal Process Mag*. 2023;40(1):85–97. <https://doi.org/10.1109/MSP.2022.3199595>.
99. Beck A, Teboulle M. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J Imaging Sci*. 2009;2(1):183–202. <https://doi.org/10.1137/080716542>.
100. Parikh N, Boyd S. Proximal algorithms. *Found Trends Optim*. 2014;1(3):127–239. <https://doi.org/10.1561/2400000003>.
101. Tassano M, Delon J, Veit T. FastDVDnet: Towards real-time deep video denoising without flow estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2020. pp. 1354–1363. <https://doi.org/10.1109/CVPR42600.2020.00143>.
102. Zhang Z, Deng C, Liu Y, Yuan X, Suo J, Dai Q. Ten-mega-pixel snapshot compressive imaging with a hybrid coded aperture. *Photonics Res*. 2021;9(11):2277–87. <https://doi.org/10.1364/PRJ.435256>.
103. Yuan X, Liu Y, Suo J, Durand F, Dai Q. Plug-and-play algorithms for video snapshot compressive imaging. *IEEE Trans Pattern Anal Mach Intell*. 2021;44(10):7093–111. <https://doi.org/10.1109/TPAMI.2021.3099035>.
104. Monod A, Delon J, Tassano M, Almansa A. Video restoration with a deep plug-and-play prior. 2022. arXiv preprint [arXiv:2209.02854](https://arxiv.org/abs/2209.02854).

105. He Y, Yao Y, Qi D, He Y, Huang Z, Ding P, et al. Temporal compressive super-resolution microscopy at frame rate of 1200 frames per second and spatial resolution of 100 nm. *Adv Photonics*. 2023;5(2):026003. <https://doi.org/10.1117/1.AP5.2.026003>.
106. Bohra P, Pham T-a, Long Y, Yoo J, Unser M. Dynamic Fourier ptychography with deep spatiotemporal priors. *Inverse Probl*. 2023;39(6):064005. <https://doi.org/10.1088/1361-6420/acca72>.
107. Li S, Wang F, Fu Z, Bian Y, Situ G. Dynamic quantitative phase imaging using deep spatial-temporal prior. *Opt Express*. 2025;33(4):7482–91. <https://doi.org/10.1364/OE.545458>.
108. Zhang K, Li Y, Zuo W, Zhang L, Van Gool L, Timofte R. Plug-and-play image restoration with deep denoiser prior. *IEEE Trans Pattern Anal Mach Intell*. 2021;44(10):6360–76. <https://doi.org/10.1109/TPAMI.2021.3088914>.
109. Luo W, Zhang Y, Feizi A, Göröcs Z, Ozcan A. Pixel super-resolution using wavelength scanning. *Light Sci Appl*. 2016;5(4):16060. <https://doi.org/10.1038/lsa.2016.60>.
110. Gao Y, Cao L. Generalized optimization framework for pixel super-resolution imaging in digital holography. *Opt Express*. 2021;29(18):28805–23. <https://doi.org/10.1364/OE.434449>.
111. Liang J, Wang LV. Single-shot ultrafast optical imaging. *Optica*. 2018;5(9):1113–27. <https://doi.org/10.1364/OPTICA.5.001113>.
112. Qi D, Zhang S, Yang C, He Y, Cao F, Yao J, et al. Single-shot compressed ultrafast photography: a review. *Adv Photon*. 2020;2(1):014003. <https://doi.org/10.1117/1.AP2.1.014003>.
113. Yuan X, Brady DJ, Katsaggelos AK. Snapshot compressive imaging: theory, algorithms, and applications. *IEEE Signal Process Mag*. 2021;38(2):65–88. <https://doi.org/10.1109/MSP.2020.3023869>.
114. Ho J, Salimans T, Gritsenko A, Chan W, Norouzi M, Fleet DJ. Video diffusion models. In: *Advances in Neural Information Processing Systems*. 2022. pp. 8633–8646. <https://doi.org/10.48550/arXiv.2204.03458>.
115. Buzzard GT, Chan SH, Sreehari S, Bouman CA. Plug-and-play unplugged: optimization-free reconstruction using consensus equilibrium. *SIAM J Imaging Sci*. 2018;11(3):2001–20. <https://doi.org/10.1137/17M1122451>.
116. Guizar-Sicairos M, Thurman ST, Fienup JR. Efficient subpixel image registration algorithms. *Opt Lett*. 2008;33(2):156–8. <https://doi.org/10.1364/OL.33.000156>.
117. Chen N, Wang C, Heidrich W. ∂H : Differentiable holography. *Laser Photonics Rev*. 2023;17(9):2200828. <https://doi.org/10.1002/lpor.202200828>.
118. Thibault P, Dierolf M, Menzel A, Bunk O, David C, Pfeiffer F. High-resolution scanning X-ray diffraction microscopy. *Science*. 2008;321(5887):379–82. <https://doi.org/10.1126/science.1158573>.
119. Maiden AM, Rodenburg JM. An improved ptychographical phase retrieval algorithm for diffractive imaging. *Ultramicroscopy*. 2009;109(10):1256–62. <https://doi.org/10.1016/j.ultramic.2009.05.012>.
120. Thibault P, Menzel A. Reconstructing state mixtures from diffraction measurements. *Nature*. 2013;494(7435):68–71. <https://doi.org/10.1038/nature11806>.
121. Monakhova K, Richter SR, Waller L, Koltun V. Dancing under the stars: video denoising in starlight. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022. pp. 16241–16251. <https://doi.org/10.1109/CVPR52688.2022.01576>.
122. Sun Y, Wu Z, Xu X, Wohlberg B, Kamilov US. Scalable plug-and-play ADMM with convergence guarantees. *IEEE Trans Comput Imaging*. 2021;7:849–63. <https://doi.org/10.1109/TCI.2021.3094062>.
123. Pont-Tuset J, Perazzi F, Caelles S, Arbeláez P, Sorkine-Hornung A, Van Gool L. The 2017 DAVIS challenge on video object segmentation. 2017. arXiv preprint [arXiv:1704.00675](https://arxiv.org/abs/1704.00675).
124. Xue T, Chen B, Wu J, Wei D, Freeman WT. Video enhancement with task-oriented flow. *Int J Comput Vis*. 2019;127:1106–25. <https://doi.org/10.1007/s11263-018-01144-2>.
125. Zhang K, Zuo W, Zhang L. FFDNet: toward a fast and flexible solution for CNN-based image denoising. *IEEE Trans Image Process*. 2018;27(9):4608–22. <https://doi.org/10.1109/TIP.2018.2839891>.
126. Chang X, Zhao R, Jiang S, Shen C, Zheng G, Yang C, et al. Complex-domain-enhancing neural network for large-scale coherent imaging. *Adv Photon Nexus*. 2023;2(4):046006. <https://doi.org/10.1117/1.APN.2.4.046006>.
127. Liu J, Sun Y, Xu X, Kamilov US. Image restoration using total variation regularized deep image prior. In: *ICASSP 2019–2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE; 2019. pp. 7715–7719. <https://doi.org/10.1109/ICASSP2019.8682856>.

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